Shopping Behaviour Analysis

Barbara Klimova, Aleem Shaikh
Brain Station - Data Analytics Certificate
22.10.2025

Agenda

Dress
to
Impress

Basic to premium clothing & accessories

I. Problem Statement
II. Hypothesis Formulation
III. Data Cleansing & Organization
IV. Analysis of Findings
V. Visualization of the Storyline
VI. Summary & Recommended Solutions

The Problem Statement

Target Audience

The Leadership Team at Dress to Impress has an interest to increase revenue and scale the business into further product lines.

Target Focus

The focus is on on evaluating seasonality and consumer trends by demographic profile based on age, gender and shopping behaviour patterns.

Target Purpose

The Head of Sales would like to...

- Understand their consumer demographic trends to establish a sales strategy for expansion of product lines
- Set promotions by seasonal trends

The Head of OPS would like to...

- Set Free Shipping on orders over certain threshold
- Determine whether AfterPay is a sought after payment method

The Head of MKT would like to...

- Send out targeted promotional EDMs to subscribers to alert on seasonal product line expansions
- Send out targeted promotional EDMs by shopping behavior per territory

Main goal for the LT is to...

- Set a marketing promotional calendar strategy for the 2026 Calendar year

Hypothesis

Customer Behaviour Hypotheses

1. Older customers tend to spend more per purchase than younger customers.
2. Subscribed customers have higher purchase frequency and total purchase count than unsubscribed ones.
3. The subscribed customers would have the highest average units per order from the seasonal items based on climate.
4. Product category preferences vary significantly by gender; females most likely purchase more accessories and footwear.

Product Performance Hypotheses

1. Products from “summer” or “spring” season show spikes in number of purchases midway through the season whilst “winter” and “fall” at the beginning of the season.
2. Higher-rated products are purchased more frequently and generate higher revenue.

Purchase Dynamics Hypotheses

1. The average customer would spend more on their purchase if they are offered free shipping.
2. Younger customers prefer digital wallets, while older customers lean toward credit cards.

Territory Trend Hypotheses

1. Customers in the Northeast and Midwest are purchasing higher amount of winter and fall clothing than customers in the South and West; purchasing higher amount of summer and spring clothing.

Data and Tools Used

Where did you find the dataset?

Our dataset was found on Kaggle.com

Did you have to join multiple datasets in order to create the ideal dataset?

No, however our dataset had to be cleaned, and variables e.g. purchase date and purchase id added so that we could build a relational database

Which tools did you use to undertake the processing and statistical analysis?

- Microsoft Excel
- SQL
- Tableau

What other data would you have liked to have in an ideal scenario?

•Lead source of acquisition channel to assist the Marketing team on strategy of most profitable channels and expansion into further channels
•Territories instead of states and Sales reps assigned to those territories

Data Processing

Data cleansing process:

- Inspected column types, nulls, and row counts across tables
- Remove duplicates using DISTINCT and ROW_NUMBER() on product attributes
- Validated joins between purchases, products, and customers
- Customers and Products are connected through Purchases.
- Standardized inconsistent values and normalized categories
- Parsed purchase_date into MONTH() and YEAR() for seasonal analysis

Determining Outliers:

- Flagged outliers in purchase amount, rating, and purchase count
- Set business rules to derive persona based on Median, STDEV to create Upper and Lower Limits (Example: Upper limit: Median + 0.5* STDEV, Lower Limits: Median - 0.5*STDEV)
- Preserved valid outliers unless clearly erroneous or duplicated

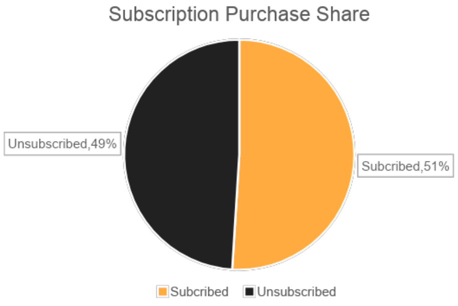
Set Assumptions:

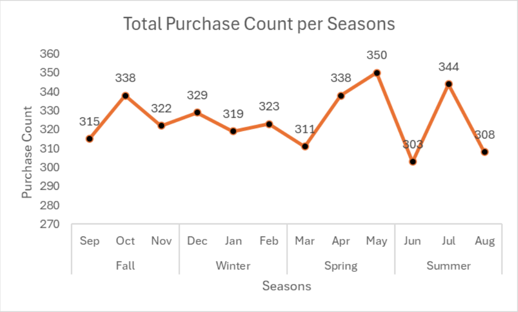
<u>Assumption 1:</u>
The dataset is suggesting that each customer only made one order, therefore limitation of following customer retention journey
<u>Assumption 2:</u>
The average purchase per customer of one type of item is 25 units, which is quite high. Data suggests the customers are purchasing the units to resell at their own store. Dress to Impress -> wholesaler
<u>Assumption 3:</u>
Our dataset from Kaggle did not include a purchase date therefore it is fictional to determine seasonality and could be skewing our data

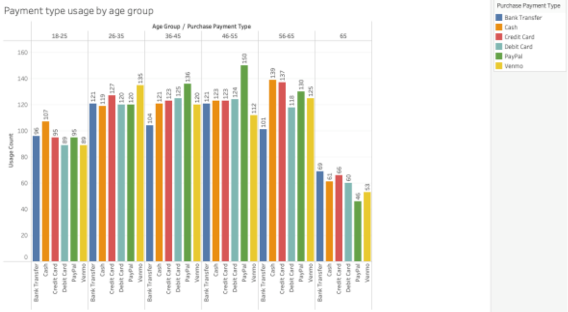
Data Processing

Were there any surprising insights from your data processing/exploration?

1. The gap between average purchase count between subscribed and unsubscribed customers was quite narrow only 2% higher for subscribed customers a opposed to unsubscribed
2. Males seem to be purchasing twice as much in every category compared to females
3. Highest rated products have not necessarily been purchased the most frequently and didn't generate higher revenue than the lower rated products.
4. The youngest customer actually prefer cash and the customers that are 65+ prefer payment by credit card







Statistical Analysis

Marketing trends analysis

I. Consumer Trends by Age group
II. Consumer Trends by Gender
III. Consumer Trends by Ratings

Operations trends analysis

I. Shipping Preferences
II. Payment type usage by Age group

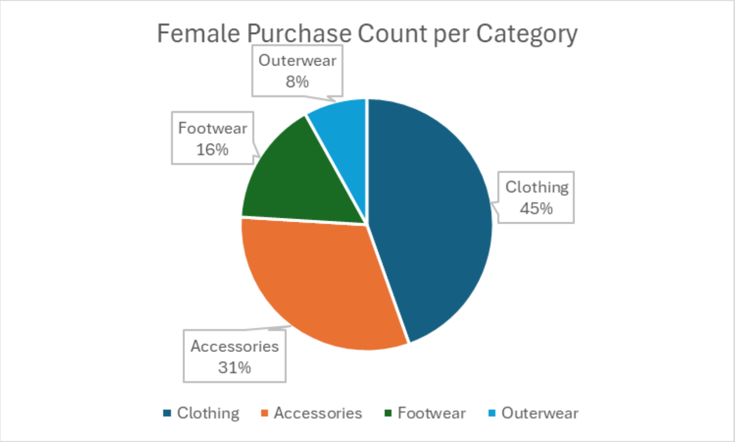
Sales trends analysis

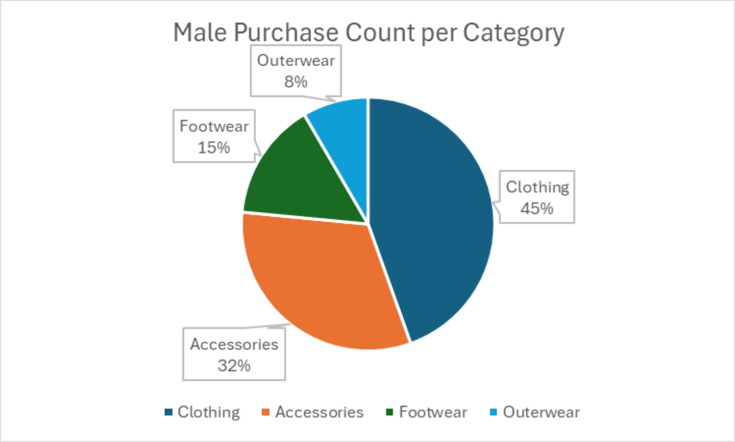
I. Subscribers purchasing trends by season
II. Sales spikes based on seasonality
III. Seasonal Trends by Territory

Do product preferences vary by gender?

It seems that either our product portfolio has much more range and/or is much more attractive for males as males seem to be purchasing twice as much in every category. Our advice to the Marketing team is to boost the targeting of the female audience and expand the product ranges for females to attract more sales from this target customer profile.

Customer_gender	Product_category	Purchase_count
Female	Clothing	556
Female	Accessories	392
Female	Footwear	199
Female	Outerwear	101
Male	Clothing	1181
Male	Accessories	848
Male	Footwear	400
Male	Outerwear	223

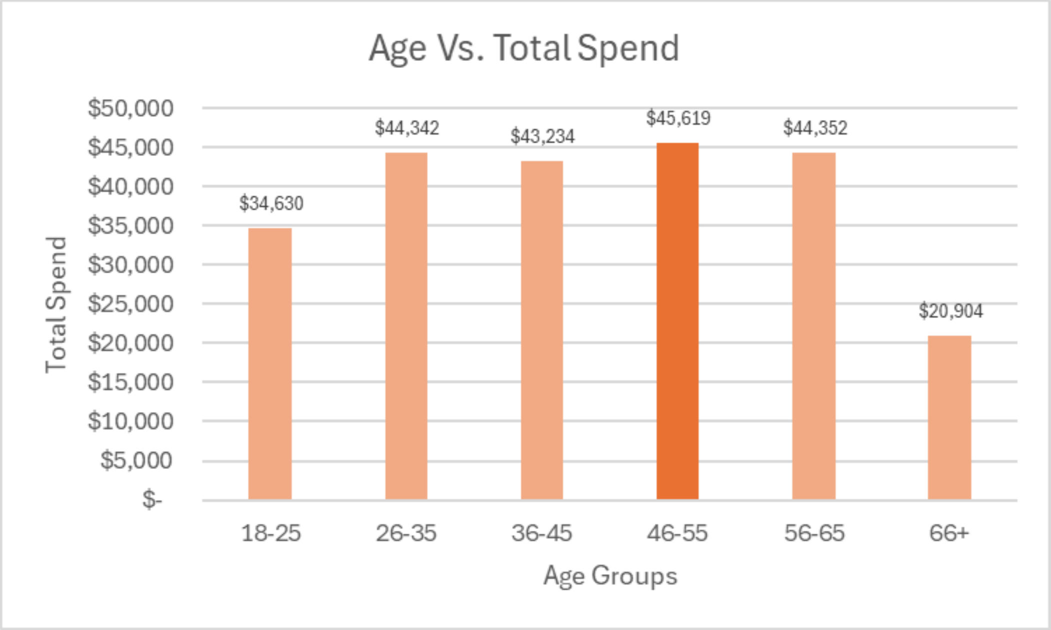




Older customers spend more per purchase than younger customers

Customers aged 46-55 have the most disposable income as they have spent the most in total purchases

Age_Group	Total_purchase_amount	
18-25	$	34,630.00
26-35	$	44,342.00
36-45	$	43,234.00
46-55	$	45,619.00
56-65	$	44,352.00
66+	$	20,904.00

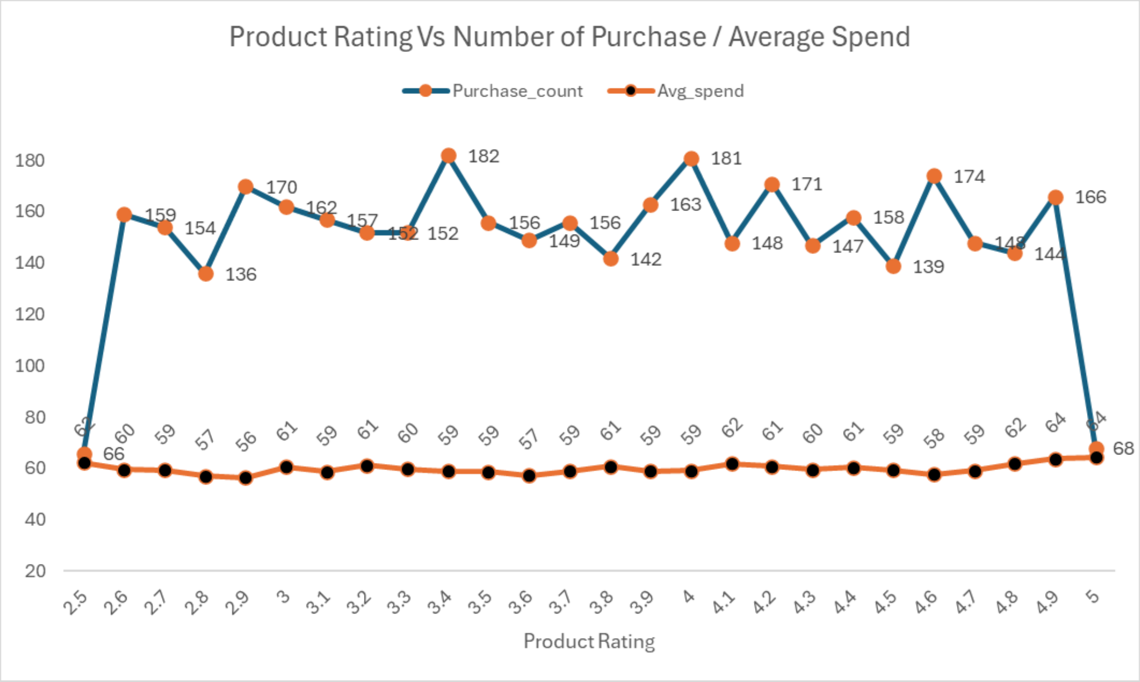


Customer Persona

- **Young Budget Shoppers**: Age < 30, Spend < Low Spend, Purchase Count > High Frequency
- **Loyal High Spenders**: Age between 36–60, Spend > High Spend, Subscribed
- **Occasional Deal Seekers**: Purchase Count < Low Frequency, Spend < Low, Unsubscribed

Personas	Customers	Revenue	ARPU
Young Budget Shoppers	168	$ 128,493	$ 764.84
Loyal High Spenders	230	$ 569,076	$ 2,474.24
Occasional Deal Seekers	858	$ 413,151	$ 481.53

Do higher-rated products sell more?



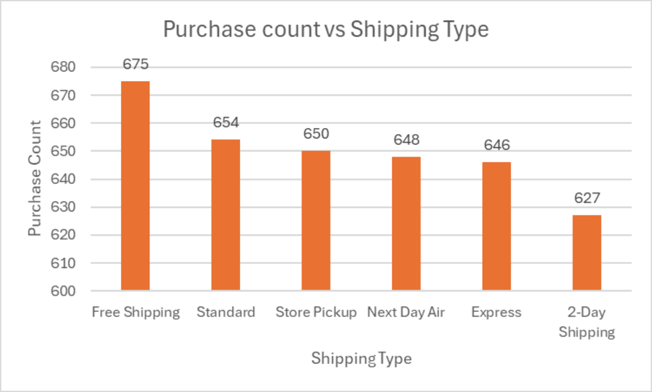
Our Hypothesis proved to be wrong as the highest rated products have not necessarily been purchased the most frequently and didn't generate higher revenue than the lower rated products. This trend is showing that when people want to buy in bulk they might not care so much about the higher rating of the product, rather looking at the price of the individual product.

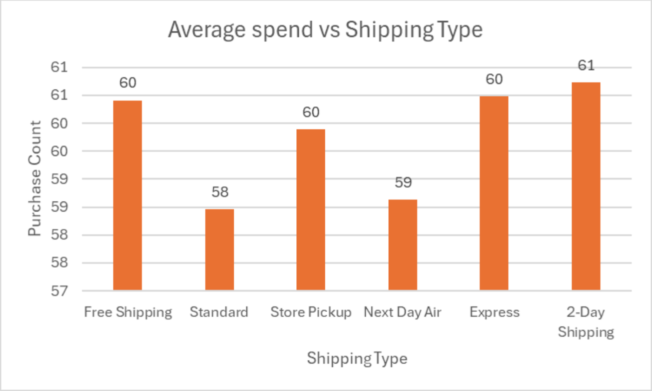
The average highest sale amount was spent on the highest rated product, which comes to show that on average customers are willing to spend more on a higher rated product at less volume.

Does shipping type affect the number of purchases?

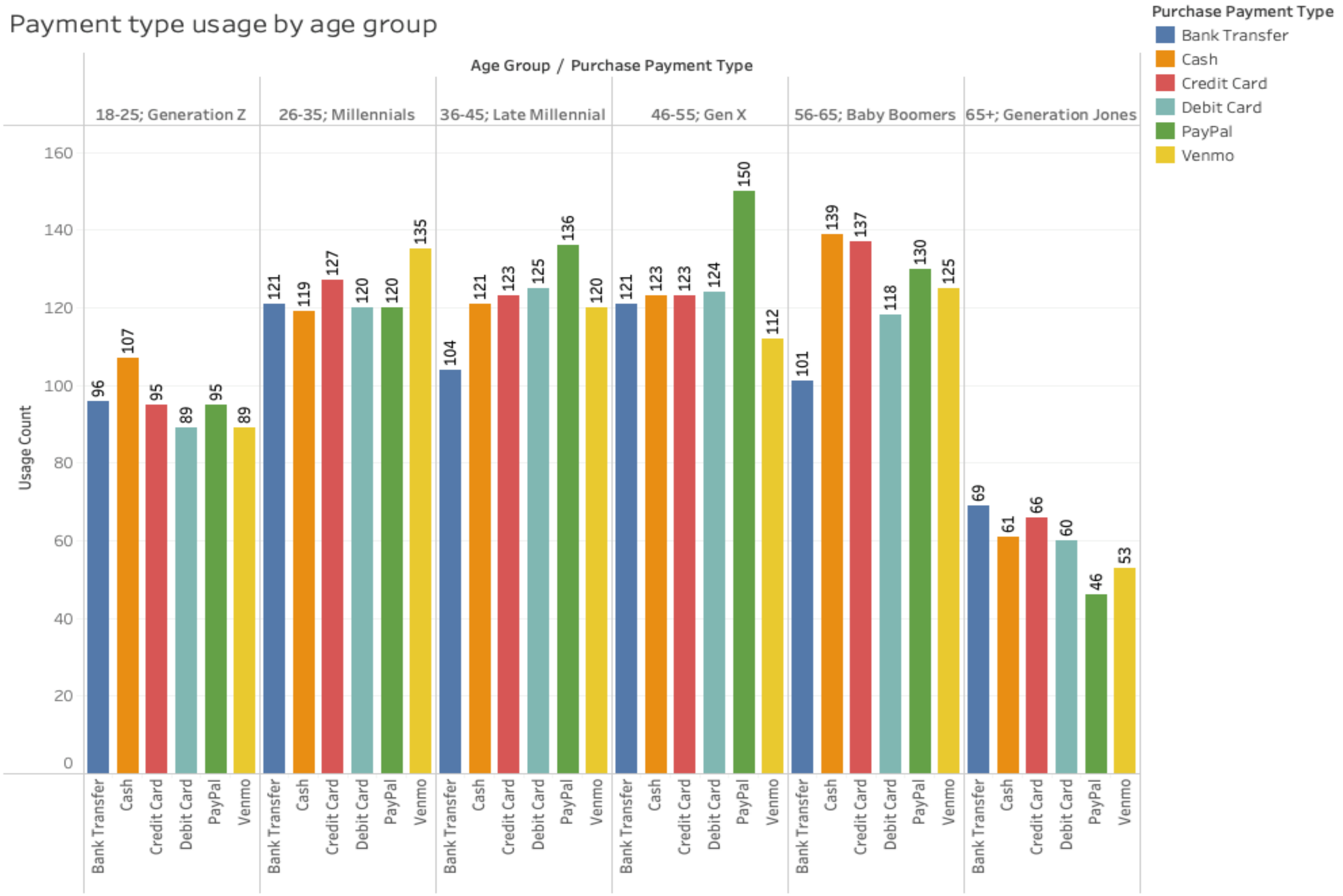
The purchase count is highest on free shipping and the average customer is willing to spend more when offered free shipping.

Purchase_shipping_type	Purchase_count	Avg_spend
Free Shipping	675	60
Standard	654	58
Store Pickup	650	60
Next Day Air	648	59
Express	646	60
2-Day Shipping	627	61

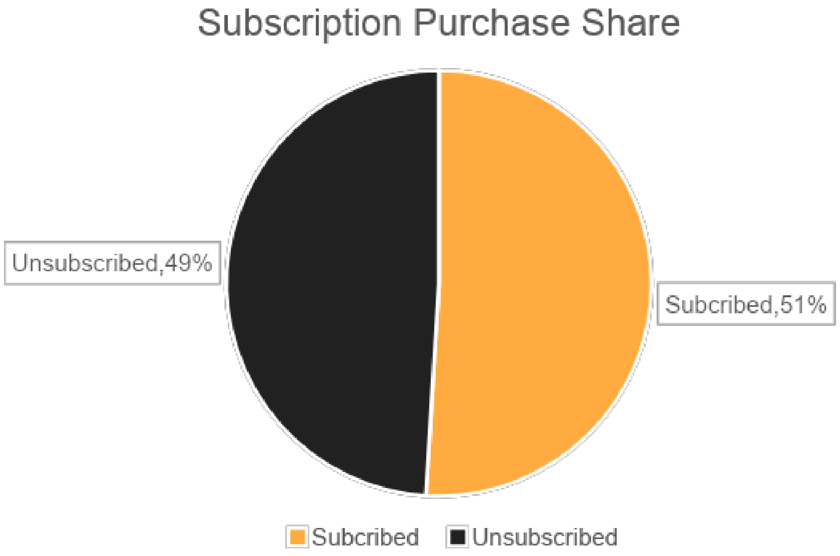




Payment type usage by age group



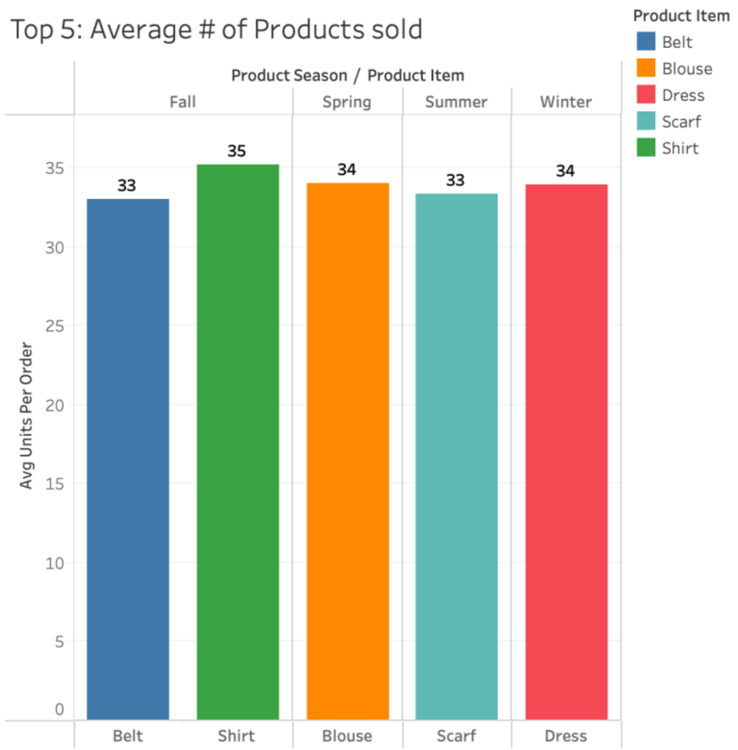
Do subscribed customers purchase more frequently?

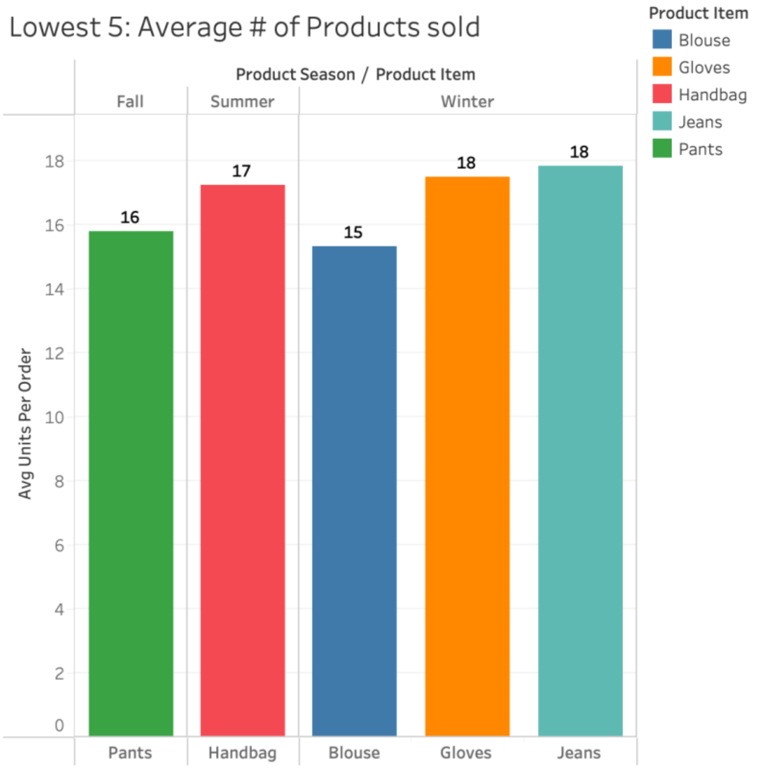


Average purchase count stands at 26.08 for subscribed customers vs. 25.08 for unsubscribed.

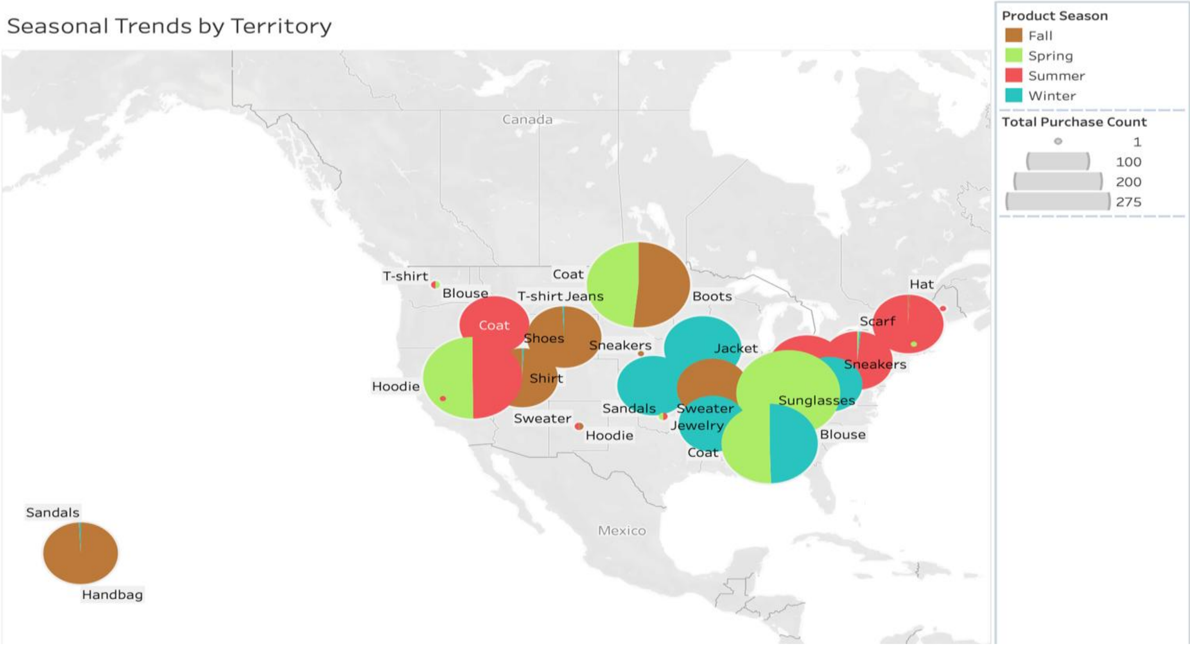
Customer_subscription	Avg_purchase_count
Subscribed	26.0845
Unsubscribed	25.0804

Subscriber seasonal purchasing patterns





Seasonal Trends by Territory



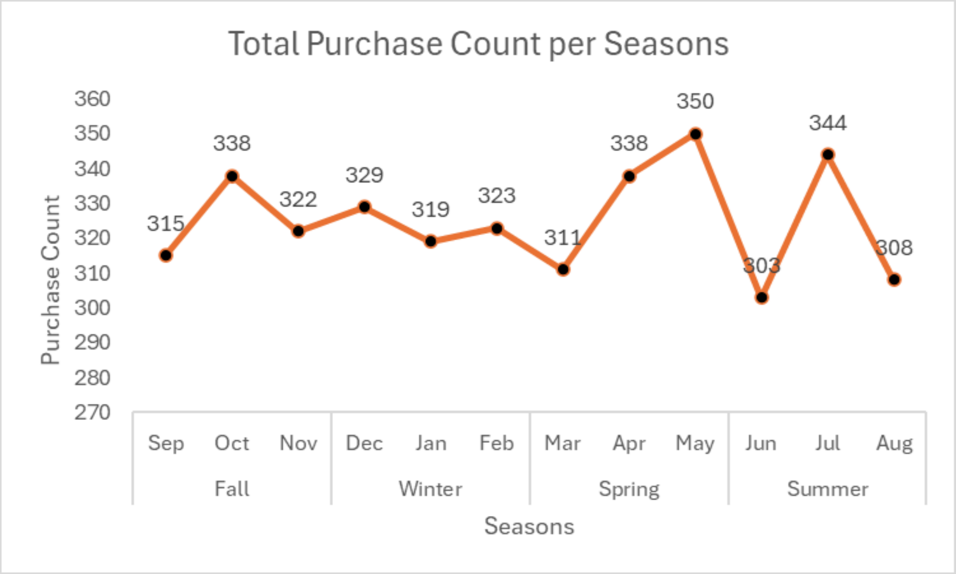
Midwest & Northeast Highest Purchases

Product Se..	Product Item	Territory: Midwest	Territory: Northeast
Fall	Boots	140	
	Shorts	130	
Spring	Coat	131	
Summer	Sneakers	151	127
	Sunglasses		125
Winter	Jacket	153	
	Jewelry	131	

South & West Highest Purchases

Product Se..	Product Item	Territory: South	Territory: West
Fall	Handbag		141
	Shirt		139
	Shorts		129
Spring	Coat	244	
	Hoodie		125
	Sweater	150	
Summer	Belt		124
	Coat		125
Winter	Blouse	117	
	Gloves	112	
	Hoodie	118	

Establishing trends in seasonal spikes



Seasons	Months	Purchase Count
Fall	Sep	315
	Oct	338
	Nov	322
Winter	Dec	329
	Jan	319
	Feb	323
Spring	Mar	311
	Apr	338
	May	350
Summer	Jun	303
	Jul	344
	Aug	308

- In Fall and Summer purchases were the highest throughout the season suggesting that most customers shop more midway through the season after potential discounts have been offered and trends have been set.
- Whereas in Spring the highest number of purchases occurred in the last month, suggestion to Marketing for promoting spring trends before the season starts to spike up the sales in the earlier month of the season.
- Lastly, winter sales were the highest in December suggesting success of strong impact from early winter campaigns and promotions.

Summary Conclusions

Was our Target Purpose met?

The Head of Sales would like to…

Target 1: Understand their consumer demographic trends to establish a sales strategy for expansion of product lines
Recommendation 1: Focus product line expansion on the 36–55 age group, who show the highest total spend, and diversify offerings for female customers, who currently purchase less across all categories. Tailor product development to align with these underserved yet high-potential segments.

Target 2: Set promotions by seasonal trends
Recommendation 2: Start with promotions one month prior to the season so that we can increase the early season sales.

The Head of OPS would like to…

Target 3: Set Free Shipping on orders over certain threshold
Recommendation 3: As a wholesaler with average sales of 26 units per item we suggest free shipping on orders with MOQ of 10 per item.

Target 4: Determine whether AfterPay is a sought after payment method

Recommendation 4: In order to motivate Early Millennials, Baby Boomers & The Generation Jones and support their preferred payment usage by increasing payment options on credit; we would recommend to offer AfterPay; buy more pay later.

The Head of MKT would like to…

Target 5: Send out targeted promotional EDMs to subscribers to alert on seasonal product line expansions

Recommendation 5: EDM to all customers notifying of the new rewards scheme providing tiered discounts for subscribers who make the highest amount of purchases

Target 6: Send out targeted promotional EDMs by shopping behavior per territory

Recommendation 6: EDM by geographic location to set trends via blogs etc. to set product incentives as per territory climate

The LT would like to…

Target 7: Set a marketing promotional calendar strategy for the 2026 Calendar year

Recommendation 7: Trends in seasonal spikes to set subscriber promotional calendar per each territory

Key Learnings

When running the queries and realising that some of the results don’t make much sense, ensure that the data has been inputted in the dataset; reporting in real life accurately.

Solution -> Cross check the data with the department manager/SMEs.

Thank You

References

1. Link to dataset from Kaggle : https://www.kaggle.com/datasets/zubairamuti/shopping-behaviours-dataset
2. Shopping Behavior updated: attached
3. Final queries file : attached
4. Individual table csvs: attached
5. Persona Statistical Analysis: attached